Computability: Exercise 7

Due: May 27^{th}, 2026

Question 1

For languages L and L', we write $L' \leq_p L$ if there exists a reduction TM from L' to L who's running time is polynomial in the length of its input.

a. Show that $HALT \leq_p NON-EMPTY$.

b. Show that A_{TM} is complete in RE with respect to polynomial-time reductions.

Question 2

Show that P is closed under concatenation and Kleene closure.
Hint: Dynamic programming is a friend

Question 3

Recall that the running time of a TM M is defined by the function $t_M(n)$, which for a parameter n returns the maximal running time of M over all inputs of length at most n (if M does not halt on such an input, define the running time to be ∞). Consider the language $L \stackrel{\text{def}}{=} \{\langle M\rangle \colon t_M(n) = O(2^n)\ \}$. Show that L is RE-hard with respect to polynomial reductions.

Question 4

In the recitation we showed a polynomial-time reduction from CLIQUE to IS. In this question you are required to show similar reductions for other pairs of problems.

a. Show that IS $\leq_p$ CLIQUE.

b. Let

$$\text{E3SAT} = \{\langle\varphi\rangle : \varphi \text{ is a satisfiable CNF formula with exactly 3 different literals in each clause}\}$$

Show that E3SAT $\leq_p$ E4SAT, where E4SAT is similar to E3SAT except that each clause contains exactly 4 distinct literals instead of 3.
Hint: You may use the following observation from logic: if both $p \vee q$ and $p \vee \overline{q}$ are true, then p must be true (verify it first...)

Question 5

Recall that in a graph G, a *path* from a vertex u to a vertex v is a sequence $x_0, \ldots, x_\ell$ such that $x_0 = u$, $x_\ell = v$ and for every $0 \leq i < \ell$ there is an edge from x_i to x_{i+1}. The length of the path is ℓ (the number of edges it visits, including repetitions). A path is called *simple* if it does not repeat any vertices.

Define:

$$\text{D-KPATH} = \left\{ \langle G, u, v, k \rangle \;:\; \begin{array}{l} G \text{ is a directed graph, and there is a simple path} \\ \text{from } u \text{ to } v \text{ of length at least } k. \end{array} \right\}$$

$$\text{U-KPATH} = \left\{ \langle G, u, v, k \rangle \;:\; \begin{array}{l} G \text{ is an undirected graph, and there is a simple path} \\ \text{from } u \text{ to } v \text{ of length at least } k. \end{array} \right\}$$

Show that U-KPATH $\leq_p$ D-KPATH.

Question 6

Prove $NP \subseteq EXP$. Deduce $NP \subseteq R$.